

Fruit Fly

Team: Cheesecake

Element F: Consideration of Design Viability

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

Our solution will address our stakeholders' problem by getting rid of the flies. Our prototype design was very promising; it's easily maintainable, lightweight, and durable. The funnel shaped insert will allow the flies to go into the trap without them being able to leave. The bottom of our design screws off as well, this allows the vinegar and dead flies to be easily cleaned out.

We know our design is going to be easy to build because we are going to 3D print it. We also made a prototype design using TinkerCAD. One thing we did to go above and beyond the design requirement was that we added the screw off the bottom to make it easy to clean. Our expert, Mr. S, said that our design needed to be able to withstand the conditions of the boys dorm. The trap needs to have a neutral odor and cant add inconvenience to the life of the residents. Mr. S is a dorm duty member and has lived on campus for years.